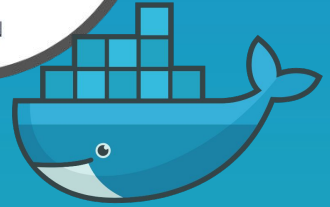




Module Checklist

Containers with Docker

By TechWorld with Nana



Video Overview

- ★ What is a Container?
- ★ Container vs Image
- ★ Docker (Container) vs Virtual Machine
- ★ Docker Architecture and its components
- ★ Main Docker Commands
- ★ Debug Commands
- ★ Demo Project: Overview
- ★ Demo Project: Developing with Docker
- ★ Demo Project: Docker Compose - Running multiple services
- ★ Demo Project: Dockerfile - Building our own Docker Image
- ★ Demo Project: Private Docker Repository - Pushing our Docker Image into a private Registry on Nexus
- ★ Demo Project: Deploying our containerized application
- ★ Docker Volumes - Persist data in Docker
- ★ Demo Project: Volumes - Configuring persistence for our application
- ★ Demo Project: Run Nexus as Docker container
- ★ Docker Best Practices

Check your progress... 1/6

What is a Container?

- ☒ Watched video

Container vs Image

- ☒ Watched video
- ☒ Installed Docker on your local machine
- ☒ **Demo executed** - run two different Versions of Postgres Docker Images

Useful Links:

- Docker Installation Guides for different OS:
<https://docs.docker.com/get-docker/>
- Postgres Docker Images: https://hub.docker.com/_/postgres

Docker vs Virtual Machine

- ☒ Watched video

Docker components

- ☒ Watched video

Check your progress... 2/6



Main Docker Commands

- ☐ Watched video
- ☐ **Demo executed**
 - ☐ Pull Redis Docker Image (docker pull)
 - ☐ List existing Docker Images (docker images)
 - ☐ Run Container (docker run)
 - ☐ Run Container in a detached mode (docker run -d)
 - ☐ List running containers (docker ps)
 - ☐ Start container (docker start)
 - ☐ Stop container (docker stop)
 - ☐ List all containers - running and stopped ones (docker ps -a)
 - ☐ Bind port (docker run -p)

Useful Links:

- Redis Docker Images: https://hub.docker.com/_/redis

Debug Commands

- ☐ Watched video
- ☐ **Demo executed**
 - ☐ See logs of container (docker logs)
 - ☐ Get interactive terminal of running container for troubleshooting (docker exec -it)

Demo Project: Overview

- ☐ Watched video

Check your progress... 3/6



Demo Project: Developing with Docker

- ☐ Watched video
- ☐ **Demo executed**
 - ☐ Git cloned example git project or created new one
 - ☐ Pulled mongodb image
 - ☐ Pulled mongo-express image
 - ☐ Created mongo-network
 - ☐ Started mongodb container with all necessary parameters
 - ☐ Started mongo-express container with all necessary parameters
 - ☐ Created new database via Mongo Express UI
 - ☐ Configured Nodejs application code to connect with database

Useful Links:

- MongoDB Docker Image: https://hub.docker.com/_/mongo
- Mongo Express Docker Image: https://hub.docker.com/_/mongo-express

Demo Project: Docker Compose - Running multiple services

- ☐ Watched video
- ☐ **Demo executed**
 - ☐ Installed Docker Compose (should already be installed with Docker Desktop)
 - ☐ Created a docker-compose file to start mongodb and mongo-express containers instead of using docker run
 - ☐ Created new database

Useful Links:

- Docker Compose Installation Guides for different OS: <https://docs.docker.com/compose/install/>

Check your progress... 4/6



Demo Project: Dockerfile - Building our own Docker Image

- ☐ Watched video
- ☒ **Demo executed**
 - ☐ Created Dockerfile for our Node application (the name of the file MUST be Dockerfile!)
 - ☐ Built Docker Image from our Dockerfile and tag it
 - ☐ Started newly created Docker Image

Demo Project: Private Docker Repository - Push/Pull to Nexus Repository

- ☐ Watched video
- ☒ **Demo executed**
 - ☐ Created a Docker Repository on Nexus
 - ☐ Created a User Role for Docker Repository on Nexus
 - ☐ Configured Repository Connector (port 8083)
 - ☐ Configured Firewall Rule to open port 8083 on Droplet
 - ☐ Configured Token Issuing on Nexus (Realm - activate Docker Bearer Token Realm)
 - ☐ Configured insecure registries for Nexus IP and Port in Docker Desktop (Docker Engine Tab)
 - ☐ Logged in to Nexus Docker Repo (docker login)
 - ☐ Pushed Docker Image to Nexus Repo
 - ☐ Fetched Docker Image from Nexus Repo

Check your progress... 5/6



Demo Project: Deploying our containerized application

- ☐ Watched video
- ☐ **Demo executed**
 - ☐ Added our example application to Docker compose file
 - ☐ Changed mongodb server url from localhost to mongodb service name in Node Code
 - ☐ Started docker containers with docker-compose

Docker Volumes - Persist data in Docker

- ☐ Watched video

Demo Project: Volumes - Configuring persistence for our application

- ☐ Watched video
- ☐ **Demo executed** - defined a Named Volume in Docker Compose File

Check your progress... 6/6



Demo: Project Run Nexus as Docker Container on DigitalOcean Droplet

- ☒ Watched video
- ☒ **Demo executed**
 - ☒ Created a new Droplet
 - ☒ Attached Droplet to existing Firewall
 - ☒ Installed Docker on Droplet
 - ☒ Created docker volume to persist Nexus data
 - ☒ Ran Nexus as Docker container with necessary parameters
 - ☒ Accessed Nexus in browser

Useful Links:

- Nexus Docker Image: <https://hub.docker.com/r/sonatype/nexus3>

More Resources...

Official Resources

- Best practices for writing Dockerfiles:
https://docs.docker.com/develop/develop-images/dockerfile_best-practices/
- Docker development best practices:
<https://docs.docker.com/develop/dev-best-practices/>
- Tips for Caching, reducing Image size, maintainability, reproducibility:
<https://www.docker.com/blog/intro-guide-to-dockerfile-best-practices/>
- **Tip:** Enforce Dockerfile best practices automatically by using a static code analysis tool (e.g. <https://github.com/hadolint/hadolint>)